Assignment #01 – Computer Networks

BS Artificial Intelligence

Q1. SNMP Overview (3 Marks)

What is SNMP?

SNMP (Simple Network Management Protocol) is an application layer protocol used to monitor and manage network devices such as routers, switches, servers, printers, and access points. It helps network administrators check the performance and status of devices from a central location.

SNMP allows administrators to detect network problems, monitor traffic, and ensure that devices are working properly. It reduces the need to physically inspect each network device.

Role of SNMP in Network Management

SNMP plays an important role in managing large networks. It helps administrators collect information about device performance, CPU usage, memory usage, bandwidth consumption, and network errors. If a problem occurs, SNMP can generate alerts so that administrators can take immediate action.

The main components of SNMP are:

SNMP Manager: A central system that monitors and controls devices.

SNMP Agent: Software running on network devices that collects information.

Managed Device: Any network device being monitored.

MIB (Management Information Base): A database containing information about network devices.

Diagram

Draw:

SNMP Manager ↔ SNMP Agent ↔ Managed Device

MIB connected to Agent

Q2. Configuration and Performance Monitoring (3 Marks)

What is Network Configuration Monitoring?

Network configuration monitoring is the process of tracking and managing the settings and configurations of network devices. It helps administrators ensure that all devices are properly configured and operating according to organizational requirements.

Configuration monitoring helps identify unauthorized changes, maintain backups, and reduce network downtime.

Why is Performance Monitoring Important?

Performance monitoring measures how efficiently a network operates. It helps detect problems before they become serious.

Benefits include:

- Detecting network bottlenecks
- Monitoring bandwidth usage
- Improving network reliability
- Reducing downtime
- Maintaining smooth communication

Performance monitoring allows organizations to provide better services to users.

Diagram

Monitoring Station

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Server ← Router ← Switch ← Client PCs

Q3. QoS Basics (3 Marks)

What is Quality of Service (QoS)?

Quality of Service (QoS) is a network technology that manages and prioritizes network traffic. It ensures that important applications receive sufficient bandwidth and network resources.

QoS is especially important for applications such as video conferencing, online gaming, VoIP calls, and live streaming.

QoS Parameters

1. Bandwidth

Bandwidth refers to the maximum amount of data that can be transmitted over a network in a given period. Higher bandwidth allows more data to travel through the network.

2. Delay

Delay is the time taken by data packets to travel from source to destination. Excessive delay can cause communication problems.

3. Jitter

Jitter is the variation in packet arrival times. High jitter causes poor audio and video quality during communication.

4. Packet Loss

Packet loss occurs when data packets fail to reach their destination. It negatively affects communication quality.

Diagram

Highest Priority → Voice Traffic

Medium Priority → Video Traffic

Lowest Priority → Normal Data Traffic

Q4. Software-Defined Networking (SDN) (3 Marks)

What is SDN?

Software-Defined Networking (SDN) is a modern networking approach that separates the control plane from the data plane. Instead of configuring each device individually, administrators can manage the entire network through a centralized controller.

SDN makes networks more flexible, programmable, and easier to manage.

Difference Between Traditional Networking and SDN

Traditional Networking

In traditional networks, each router and switch makes its own forwarding decisions. Management becomes difficult in large networks.

SDN Networking

In SDN, a centralized controller controls network behavior. This simplifies management and improves network automation.

Advantages of SDN

- Centralized control
- Easy management
- Better scalability
- Increased flexibility
- Faster network deployment

Diagrams

Traditional Network

Router ↔ Switch ↔ Devices

(Control and Data Plane Together)

SDN Architecture

Application Layer

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Control Layer (SDN Controller)

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Infrastructure Layer (Switches and Routers)

Q5. Cloud Networking (3 Marks)

What is Cloud Networking?

Cloud networking is the use of cloud-based infrastructure and services to manage network resources. Instead of relying entirely on physical hardware, organizations can use cloud servers and virtual resources.

Cloud networking allows users to access applications, storage, and services through the internet from anywhere in the world.

Advantages of Cloud Networking

1. Scalability

Resources can be increased or decreased according to demand.

2. Cost Efficiency

Organizations save money because they do not need expensive hardware.

3. Accessibility

Users can access services from any location with an internet connection.

Additional Advantages

- Easy maintenance
- High availability
- Better disaster recovery

Diagram

Users

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Internet

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Cloud Server

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Data Center

Q6. IoT Networking (5 Marks)

What is IoT Networking?

IoT (Internet of Things) networking refers to the connection of physical devices through the internet so they can collect, exchange, and process data automatically.

IoT devices include sensors, smart lights, smart thermostats, security cameras, smart watches, and many other connected devices.

How IoT Devices Communicate in a Smart Environment

In a smart environment, sensors collect data such as temperature, humidity, motion, or light levels. This data is transmitted through a Wi-Fi router to cloud servers where it is processed and analyzed.

Users can then access the information through mobile applications and control devices remotely.

For example, in a smart home, a user can control lights, air conditioners, and security cameras through a smartphone application.

Benefits of IoT Networking

- Automation
- Remote monitoring
- Improved efficiency
- Energy savings
- Better decision making

Diagram

Sensors

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Smart Devices

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Wi-Fi Router

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Cloud Connection

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Mobile User

Q7. QoS in Real-Time Applications (5 Marks)

Real-time applications require continuous and smooth data transmission. Examples include online gaming, video conferencing, and live streaming.

Online Gaming

Online games require low delay and low packet loss. Even small delays can affect gameplay and user experience.

Video Conferencing

Applications like Zoom and Google Meet require stable bandwidth and low jitter. High delay can cause voice and video synchronization issues.

Live Streaming

Live streaming services need consistent data transmission to avoid buffering and interruptions.

Impact of QoS Parameters

Delay

High delay causes slow communication and response times.

Packet Loss

Packet loss results in missing information and reduced quality.

Jitter

High jitter causes choppy audio and video communication.

QoS prioritizes important traffic to ensure these applications work efficiently.

Diagram

High QoS Network → Smooth Communication

Low QoS Network → Delays, Jitter, Packet Loss

Q8. SDN and Cloud Networking Integration (5 Marks)

SDN and cloud networking work together to create efficient and flexible network environments.

Centralized Control

SDN provides centralized network management through an SDN controller, making administration easier.

Network Flexibility

Network configurations can be changed quickly according to business needs.

Traffic Management

SDN intelligently manages traffic flow and optimizes bandwidth usage.

Scalability

Cloud environments grow rapidly. SDN allows easy expansion without complex manual configurations.

Benefits of SDN in Cloud Environments

- Improved resource utilization
- Better performance
- Automated management
- Enhanced security
- Simplified network control

Diagram

Users/Clients

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Switches

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SDN Controller

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Virtual Servers

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Cloud Infrastructure

Bonus Question (+5 Marks)

How IoT, SDN, and Cloud Networking Work Together in a Smart City

A smart city uses IoT devices, SDN technology, and cloud networking to improve public services and city management.

IoT devices such as traffic sensors, smart streetlights, environmental sensors, and surveillance cameras continuously collect data from different areas of the city.

This data is transmitted through the network and managed by an SDN controller, which intelligently directs traffic and optimizes communication.

The collected information is stored and processed in cloud servers where advanced analytics are performed.

Citizens access services through mobile applications and web portals.

Benefits of Smart City Integration

- Reduced traffic congestion
- Improved public safety
- Better energy management
- Environmental monitoring
- Efficient public services

Smart City Diagram

Traffic Sensors

Smart Lights

Security Cameras

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Internet

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SDN Controller

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Cloud Servers

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User Applications

⭐ **Exam Important Topics from Assignment #01**

1. SNMP Manager, Agent, MIB
2. QoS Parameters (Bandwidth, Delay, Jitter, Packet Loss)
3. SDN Architecture Layers
4. Traditional Network vs SDN
5. Cloud Networking Advantages
6. IoT Components
7. Smart Home IoT Network
8. QoS for Video Conferencing
9. SDN Benefits
10. Smart City Integration (IoT + SDN + Cloud)

These are the topics most likely to appear in exams and viva questions.